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Procedural Clinical Complications, Case-Fatality Risks, and Risk Factors in Endovascular and Neurosurgical Treatment of Unruptured Intracranial Aneurysms

A Systematic Review and Meta-analysis

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Key Points

Question

What is the 30-day clinical complication risk and case-fatality rate of endovascular treatment and neurosurgical treatment of unruptured intracranial aneurysms?

Findings

In this systematic review and meta-analysis of 114 studies and 106 433 patients, among the 74 studies of endovascular treatment, the risk of procedural clinical complications was 4.96% (95% CI, 4.00%-6.12%), and the case-fatality rate was 0.30% (95% CI, 0.20%-0.40%). In 54 studies of neurosurgical treatment, the pooled complication risk was 8.34% (95% CI, 6.25%-11.10%) and the case-fatality rate was 0.10% (95% CI, 0.00%-0.20%).

Meaning

The complication risks were particularly dependent on detailed and standardized recording of complications, method of outcome assessment, and region and varied according to several patient-level, aneurysm-level, and treatment-associated risk factors.

This systematic review and meta-analysis assesses the rates of procedural complications and fatalities within the 30 days after endovascular or neurosurgical treatment for unruptured intracranial aneurysms.

Abstract

Importance

The risk of procedural clinical complications and the case-fatality rate (CFR) from preventive treatment of unruptured intracranial aneurysms varies between studies and may depend on treatment modality and risk factors.

Objective

To assess current procedural clinical 30-day complications and the CFR from endovascular treatment (EVT) and neurosurgical treatment (NST) of unruptured intracranial aneurysms and risk factors of clinical complications.

Data Sources

We searched PubMed, Excerpta Medica Database, and the Cochrane Database for studies published between January 1, 2011, and January 1, 2017.

Study Selection

Studies reporting on clinical complications, the CFR, and risk factors, including 50 patients or more undergoing EVT or NST for saccular unruptured intracranial aneurysms after January 1, 2000, were eligible.

Data Extraction and Synthesis

Per treatment modality, we analyzed clinical complication risk and the CFR with mixed-effects logistic regression models for dichotomous data. For studies reporting data on complication risk factors, we obtained risk ratios (RRs) or odds ratios (ORs) with 95% CIs and pooled risk estimates with weighted random-effects models.

Main Outcomes and Measures

Clinical complications within 30 days and the CFR.

Results

We included 114 studies (106 433 patients with 108 263 aneurysms). For EVT (74 studies), the pooled clinical complication risk was 4.96% (95% CI, 4.00%-6.12%), and the CFR was 0.30% (95% CI, 0.20%-0.40%). Factors associated with complications from EVT were female sex (pooled OR, 1.06 [95% CI, 1.01-1.11]), diabetes (OR, 1.81 [95% CI, 1.05-3.13]), hyperlipidemia (OR, 1.76 [95% CI, 1.3-2.37]), cardiac comorbidity (OR, 2.27 [95% CI, 1.53-3.37]), wide aneurysm neck (>4 mm or dome-to-neck ratio >1.5; OR, 1.71 [95% CI, 1.38-2.11]), posterior circulation aneurysm (OR, 1.42 [95% CI, 1.15-1.74]), stent-assisted coiling (OR, 1.82 [95% CI, 1.16-2.85]), and stenting (OR, 3.43 [95% CI, 1.45-8.09]). For NST (54 studies), the pooled complication risk was 8.34% (95% CI, 6.25%-11.10%) and the CFR was 0.10% (95% CI, 0.00%-0.20%). Factors associated with complications from NST were age (OR per year increase, 1.02 [95% CI, 1.01-1.02]), female sex (OR, 0.43 [95% CI, 0.32-0.85]), coagulopathy (OR, 2.14 [95% CI, 1.13-4.06]), use of anticoagulation (OR, 6.36 [95% CI, 2.55-15.85]), smoking (OR, 1.95 [95% CI, 1.36-2.79]), hypertension (OR, 1.45 [95% CI, 1.03-2.03]), diabetes (OR, 2.38 [95% CI, 1.54-3.67]), congestive heart failure (OR, 2.71 [95% CI, 1.57-4.69]), posterior aneurysm location (OR, 7.25 [95% CI, 3.70-14.20]), and aneurysm calcification (OR, 2.89 [95% CI, 1.35-6.18]).

Conclusions and Relevance

This study identifies risk factors for procedural complications. Large data sets with individual patient data are needed to develop and validate prediction scores for absolute complication risks and CFRs from EVT and NST modalities.

Introduction

The prevalence of saccular unruptured intracranial aneurysms (UIAs) in the general population is 3%.¹ Owing to the rising availability and quality of brain imaging, the number of incidentally discovered UIAs is increasing.^{2,3} Many UIAs remain asymptomatic, but some rupture, causing

subarachnoid hemorrhage (SAH). This is a subtype of stroke with a poor prognosis (a case-fatality rate of approximately 35%), often affecting relatively young patients (mean age, 60 years).⁴ Preventive treatment of UIAs, either by endovascular treatment (EVT) or neurosurgical treatment (NST), can decrease the risk of SAH, but both treatment modalities carry a risk of serious complications.⁵ Currently, the decision to treat UIAs is a balance of risk of rupture, risk of treatment complications, life expectancy, and level of patient anxiety. For estimations of the rupture risk, prediction models are available that provide absolute risks of rupture for the next 5 years based on a few easily available risk factors.^{6,7} Such robust data are lacking for the estimation of complication risk from UIA treatment.^{5,8} The best available evidence comes from meta-analyses published in 2012 and 2013 on the procedural morbidity and case-fatality risk of EVT and NST.^{9,10} However, since the publication of these reviews, there has been a further shift toward EVT as the predominant treatment modality, with an increasing variety of advanced endovascular methods being used, such as stent-assisted or balloon-assisted coiling, flow-diverting stents, and Woven EndoBridge (WEB) Aneurysm Embolization devices. In addition, previous meta-analyses did not focus on risk factors for complications apart from subgroup analyses. Our aim is to provide an overview of the recent literature on EVT and NST, with several new focuses. In addition to assessing the procedural 30-day clinical complication and case-fatality risks of both treatment modalities, we conducted a meta-analysis of the available risk factor data for clinical complications from both EVT and NST, and we separately assess the complication risk of advanced endovascular methods that were increasingly applied in clinical practice over the last few years.

Methods

Search Strategy and Selection Criteria

We systematically searched PubMed, Excerpta Medica database, and the Cochrane Database between January 1, 2011, and January 1, 2017, to retrieve all relevant articles on procedural clinical complications and case-fatality rates from EVT and NST of UIAs. A detailed query is given in eTable 1 in the [Supplement](#). We checked related articles given on PubMed and reference lists of retrieved articles for further eligible publications and compared the list of articles found with a database of references from one of us (G.J.E.R.). We performed our systematic review and meta-analysis according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) recommendations and Meta-analysis of Observational Studies in Epidemiology (MOOSE) guidelines.^{11,12}

Selection of Studies and Data Extraction

Articles were eligible for inclusion if they met the following inclusion criteria: (1) used a longitudinal design documenting procedural clinical complications and/or a case-fatality rate; (2) included at least 50 patients 18 years and older undergoing elective EVT (standard coiling or one of the following advanced endovascular methods: stent-assisted or balloon-assisted coiling, use of stents or flow diverting stents or use of WEB devices) or NST (including only clipping) between January 1, 2000, and January 1, 2017; (3) was written in Spanish, Portuguese, French, Italian, English, German, Dutch, or Scandinavian; (4) included crude or adjusted effect estimates with corresponding 95% CIs for risk factors of clinical complications available or retrievable from the data; and (5) included patients with saccular UIAs. We allowed up to 10% of the aneurysms per study to be fusiform or dissecting; up to 25% of aneurysms to be symptomatic, rather than ruptured; and up to 5% of the aneurysms to be included for retreatment. We excluded (1) animal studies; (2) studies reporting on aneurysms associated with arteriovenous malformations or in populations with specific diseases (such as collagen disorders, Moyamoya disease or syndrome, dwarfism, or autoimmune disorders); (3) studies in which previously ruptured aneurysms could not be distinguished from additional UIAs; and (4) studies in which treatment outcome was not reported separately for ruptured and unruptured aneurysms.

We predefined procedural clinical complications as treatment complications that resulted in transient or permanent morbidity or mortality and occurred during or within 30 days after the procedure. In our primary outcome measure, we included both intracranial clinical ischemic (transient ischemic attack or ischemic stroke) and hemorrhagic (intracerebral hemorrhage, subdural or epidural hematoma, and intraoperative rupture) complications, as well as unspecified complications that resulted in a deterioration in clinical outcome (worsening of modified Rankin Scale or Glasgow Outcome Scale scores or designations as unfavorable or poor, if no standardized outcome scale was used). We defined the case-fatality rate as all deaths that occurred during or within 30 days after the procedure. The assessment of outcome was based on medical records (in the case of single-center or multicenter studies) or administrative data from databases using *International Classification of Diseases, Ninth Edition (ICD-9)* and *Tenth Edition (ICD-10)* records to identify patients. For each study, we extracted details of the exact type of outcome and whether neurological deterioration was assessed with standardized outcome scales (modified Rankin Scale or Glasgow Outcome Scale).

One author (A.M.A.) performed the search and completed data extraction forms for full-text versions of the articles, including a quality assessment (Newcastle Ottawa Scale [NOS]).¹³ A second author (A.L.) validated 10% of the extraction forms. Since the level of agreement between these 2 readers was very high (100% for extraction of inclusion and exclusion criteria and 93% for the scoring of NOS forms), we refrained from double reading of the remainder of the studies. In cases of doubt, a consensus meeting was held with 2 other authors (M.D.I.V. and G.J.E.R.). If necessary, we asked authors for additional unpublished data.

For each included study we extracted (1) study characteristics: the enrollment period and midyear of treatment (median of the period during which the study was conducted), size of the study population, follow-up duration, outcome assessment (medical records vs *ICD*-based administrative databases), and scales used (modified Rankin Scale, Glasgow Outcome Scale, or others); (2) patient and aneurysm characteristics: the mean or median age of the cohort, sex distribution, aneurysm size (maximum dome diameter), and aneurysm location; (3) treatment characteristics: the modality used (EVT or NST; if EVT, standard coiling or advanced endovascular method), and (4) from studies reporting on risk factors, risk estimates per given risk factor and adjustment factors. We assessed the number of patients with clinical complications. If more than 1 category of morbidity was reported and there was no overlap, we extracted complications from both categories. Otherwise, we extracted complication data from the largest category reported. For case-fatality analyses, we assessed the number of patients who died during or within 30 days after the procedure.

Statistical Analysis

We performed separate analyses for EVT and NST and analyzed complication risk per patient. For each included study, we calculated the proportions of several patient, aneurysm, and treatment characteristics and assessed the occurrence of (1) complications causing any morbidity, including both fatal and nonfatal complications and (2) the occurrence of deaths separately, during or within 30 days after the procedure. We used mixed-effects logistic regression models for dichotomous data for the meta-analysis of proportions. Heterogeneity was classified as moderate ($I^2 = 25\%-50\%$), substantial ($I^2 = 50\%-75\%$) or considerable ($I^2 \geq 75\%$). Owing to the degree of heterogeneity found, we used random-effect models for all analyses. We performed a sensitivity analysis according to the type of outcome (ischemic or hemorrhagic intracranial clinical complications). To assess potential sources of heterogeneity across studies, we performed predefined subgroup analyses according to methodological quality (high-quality studies defined as ≥ 7 points on the NOS) and use of advanced endovascular methods. We did additional subgroup analyses according to the method of outcome assessment (medical records vs *ICD*-based administrative databases, the use of standardized outcome scales [yes or no], region [Europe, North-America, Asia, or other], and midyear of treatment [periods divided into tertiles]). To further study the influence of time on outcome, we performed meta-regression analyses using the midyear of each study period to express the percentage change of the crude complication risk or case-fatality rate per year. For each of the studies reporting data on risk factors of complications, we obtained risk ratios (RRs) or odds ratios (ORs) with 95% CIs or the raw patient numbers for each risk factor. We used the most adjusted estimate per study. If definitions or cutoff values of risk factors and treatment outcome allowed harmonization in comparable risk factor groups, we subsequently pooled ORs or RRs with a generic inverse variance-weighted, random-effects model.

Results

In total, 5423 articles were screened, of which 114 articles [14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100,101,102,103,104,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119,120,121,122,123,124,125,126,127](#) met the eligibility criteria for this review (eFigure 1 and eTables 2-4 in the [Supplement](#)). For EVT, we included 74 studies [16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89](#) that included 71 819 patients with 73 066 aneurysms (eTable 3 in the [Supplement](#)) and for NST, we included 54 studies [24,30,32,37,38,57,64,65,68,77,87,90,91,92,93,94,95,96,97,98,99,100,101,102,103,104,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119,120,121,122,123,124,125,126,127,128,129](#) with a total of 34 614 patients with 35 197 aneurysms (eTable 4 in the [Supplement](#)). Fourteen [24,30,32,37,38,39,45,47,57,64,65,68,77,87](#) of 114 studies (12.3%), with a total of 33 676 patients, reported on both EVT and NST.

Study Characteristics

Details of the included studies are given in [Table 1](#) and eTables 3 and 4 in the [Supplement](#). Most of the studies had a retrospective design (68 EVT studies [16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,34,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,72,73,74,75,76,77,78,79,80,81,82,83,85,86,87,128](#) [92%] with 71 098 patients; 53 NST studies [18,24,30,32,37,38,39,47,57,64,65,68,77,87,88,89,90,91,92,93,94,95,96,97,98,99,100,101,102,103,104,105,106,107,108,109,110,111,112,113,114,115,116,117,119,120,121,122,123,124,125,126,127](#) [98%] with 34 543 patients) and were single-center cohorts (52 EVT studies [16,19,20,21,22,23,25,27,28,29,32,34,35,36,38,39,41,44,45,46,47,48,50,51,52,54,56,59,60,61,63,64,66,67,68,70,72,74,75,76,77,78,79,80,81,82,83,85,86,129](#) [70%] with 14 444 patients; 40 NST studies [18,32,38,39,47,57,64,65,68,77,87,88,91,92,93,94,95,98,99,100,102,103,105,106,107,108,109,110,111,112,113,114,116,117,118,119,120,121,122,123,124,125,126,127](#) [74%] with 9589 patients; [Table 1](#)). Of the 114 studies, 15 EVT studies [16,20,31,35,44,45,50,55,62,64,69,71,73,75,84](#) (20%) with 10 412 patients and 11 NST studies [64,77,95,102,110,112,113,118,119,120,127](#) (18%) with 4059 patients were of high methodological quality. Among 74 EVT studies, 59 [16,18,19,20,22,23,25,27,28,29,31,32,33,34,35,36,38,40,41,42,43,44,45,47,48,49,50,51,53,54,55,56,58,60,61,62,63,64,65,66,67,68,69,70,71,72,74,75,76,77,78,79,80,81,82,83,84,86,129](#) (80%) with 16 000 patients reported separately on intracranial ischemic complications and

[58,60,61,62,63,64,65,66,67,68,69,70,72,74,75,78,79,80,82,83,84,86,129](#) (78%) with 18 520 patients on hemorrhagic complications, and 17 [16,18,19,23,31,33,40,43,46,50,54,56,58,59,70,81,129](#) (23%) with 2248 patients reported on advanced endovascular methods only. Among 54 NST studies, ischemic complications were reported separately in 33 studies [18,32,38,47,64,65,68,77,99,90,91,93,94,95,97,101,103,104,105,106,107,108,109,111,112,114,115,121,122,123,124,125,127](#) (61%) with 12 691 patients and hemorrhagic complications in 36 studies [18,32,38,47,64,65,68,88,90,92,93,94,99,102,103,104,105,106,107,108,109,110,112,113,114,115,119,120,121,122,123,125](#) (67%) with 10 545 patients. Most studies originated from Asia (38 EVT studies [18,19,20,21,22,23,25,26,27,29,35,36,38,39,41,42,45,47,48,54,55,56,59,60,61,62,63,67,69,70,73,74,75,76,78,79,80](#) [51%] with 18 942 patients and 29 NST studies [18,38,39,47,88,91,92,93,94,99,100,102,105,106,108,109,110,111,112,113,114,118,119,120,121,122,125,127](#) [54%] with 7870 patients) and North America (25 EVT studies [1,4,11,16,22,25,26,27,28,33,41,47,48,53,57,58,59,61,62,63,130,131,132,133,134](#) [34%] with 51 296 patients and 15 NST studies [24,30,32,37,57,65,77,87,89,90,96,101,115,117,124](#) [28%] with 25 247 patients). Data on patient and aneurysm characteristics were available for a subset of studies ([Table 1](#)).

Table 1. Baseline Characteristics of Included Studies for Endovascular Treatment and Neurosurgical Treatment.

Characteristic	No. (%)	
	EVT	NST
Total		
Included studies ^a	74	54
Patients	71 819	34 614
Aneurysms	73 066	35 197
Procedural clinical complications	4995	6501
Fatalities	379	156
Study characteristics		
Midyear, mean (range)	2008 (2001-2014)	2008 (2001-2014)
Retrospective design	68 (92)	53 (98)
Single-center, medical records-based	52 (70)	40 (74)
Multicenter, medical records-based	12 (16)	3 (6)
ICD-based administrative database	10 (14)	11 (20)
Methodological quality		
Newcastle Ottawa Scale, points ^b		
≥7	15 (20)	11 (20)
<7	59 (80)	43 (80)
Studies reporting a standardized outcome scale ^c	40 (54)	26 (48)
Region		
Europe	9 (12)	9 (17)
North America	25 (34)	15 (28)
Asia	38 (51)	29 (54)
>1	2 (3)	1 (2)
Studies reporting risk factor data	43 (58)	26 (48)
Patient characteristics		

Characteristic	No. (%)	
	EVT	NST
Age, mean, y	57.8	56.1
Female, mean %	74.3	68.9
Aneurysm characteristics		
Size, mean, mm	7.2	6.1
Anterior circulation, mean %	84.9	97.1
Endovascular treatment modality		
Standard coil or mix ^d	57 (77)	NA
Advanced method ^e	17 (23)	NA

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Abbreviations: EVT, endovascular treatment; ICD, *International Classification of Diseases*; NA, not applicable; NVT, neurosurgical treatment; WEB, Woven EndoBridge.

^a Numbers of included studies for EVT and clipping do not add up to 114 because 14 studies reported on both treatment modalities.

^b Studies were considered high quality if they scored 7 or more points and low quality if they scored fewer than 7 points.

^c Reporting neurological outcome using the modified Rankin scale, the Glasgow Outcome scale, or another standardized outcome scale.

^d Studies did not report which endovascular treatment methods were used.

^e Studies reported on outcome following advanced endovascular methods (ie, stent-assisted coiling, balloon-assisted coiling, flow-diverting stents, or Woven EndoBridge devices).

Outcomes After EVT

The pooled crude procedural risk from EVT was 4.96% for any clinical complication (95% CI, 4.00%-6.12%; 74 studies^{[16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,128,129](#)}; 4995 complications; [Table 2](#)), 2.82% for ischemic complications (95% CI, 2.29%-3.47%; 59 studies^{[16,18,19,20,22,23,25,27,28,29,31,32,33,34,35,36,38,40,41,42,43,44,45,47,48,49,50,51,53,54,55,56,58,6](#)}

[0,61,62,63,64,65,66,67,68,69,70,71,72,74,75,76,77,78,79,80,81,82,83,84,86,129](#); 437 complications), and 0.90% for hemorrhagic complications (95% CI, 0.64%-1.27%; 58 studies [16,18,19,20,21,22,23,25,26,27,28,29,31,32,33,34,35,36,38,40,41,42,43,44,45,47,48,49,50,51,52,53,54,55,56,58,60,61,62,63,64,65,66,67,68,69,70,72,74,75,78,79,80,82,83,84,86,129](#); 212 complications). The case-fatality rate was 0.30% (95% CI, 0.20%-0.40%; 71 studies [16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,86,87,129](#); 379 deaths).

Table 2. Procedural Clinical Complication and Case-Fatality Rates From Endovascular Treatment for All Included Studies and According to Several Subgroups.

Study Characteristic	Studies, No.	Events, No.	Patients, No.	Pooled Crude Risk, % (95% CI)	I^2 , %
All studies					
Procedural clinical complications	74	4995	71 819	4.96 (4.00-6.12)	97.8
Ischemic complications	59	437	16 000	2.82 (2.29-3.47)	75.9
Hemorrhagic complications	58	212	18 520	0.90 (0.64-1.27)	76.7
Case-fatality rate	71	379	57 550	0.30 (0.20-0.40)	81.3
High-quality studies					
Procedural clinical complications	15	445	10 412	4.30 (2.59-7.07)	95.6
Case-fatality rate	14	27	9352	0.12 (0.02-0.63)	84.4
Procedural clinical complications					
Method of outcome assessment					
Medical records	64	1005	23 073	4.42 (3.49-5.59)	91.9
ICD-coded databases	10	3990	48 746	8.91 (6.38-12.31)	99.1
Standardized outcome scale ^a					
Yes	40	743	18 104	4.60 (3.50-6.05)	92.1
No	34	4252	53 715	5.39 (3.88-7.45)	98.9
Region					
Europe	9	105	1140	7.16 (4.57-11.10)	75.9
North America	25	4234	51 296	8.26 (6.31-10.70)	98.5
Asia	38	637	18 942	3.31 (2.46-4.44)	91.4
>1 Region	2	19	361	1.65 (0.80-2.50)	63.2
Period					

Study Characteristic	Studies, No.	Events, No.	Patients, No.	Pooled Crude Risk, % (95% CI)	I^2 , %
2001-2007	23	2414	29 280	4.81 (3.00-7.62)	98.9
2008-2010	27	1190	20 486	5.13 (4.03-6.51)	90.1
2011-2014	20	1363	21 652	4.64 (2.87-7.43)	97.7
Period, case-fatality data					
2001-2007	23	134	29 280	0.32 (0.20-0.50)	62.6
2008-2010	25	25	7277	0.16 (0.06-0.47)	69.8
2011-2014	19	217	20 592	0.23 (0.09-0.62)	80.9
Advanced endovascular methods ^b					
Procedural clinical complications	17	189	2248	6.13 (4.29-8.70)	78.9
Case fatalities	17	14	2248	0.43 (0.17-1.10)	36.4

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Abbreviation: *ICD, International Classification of Diseases.*

^aReporting neurological outcome using the modified Rankin Scale, the Glasgow Outcome Scale, or another standardized outcome scale.

^bAdvanced endovascular methods (ie, stent-assisted coiling, balloon-assisted coiling, flow-diverting stents or Woven EndoBridge devices).

Subgroup Analyses

Among 15 high-quality EVT studies,^{16,20,31,35,44,45,50,55,62,64,69,71,73,75,84} the complication risk was 4.30% (95% CI, 2.59%-7.07%; 445 complications; [Table 2](#)) and the case-fatality rate was 0.12% (95% CI, 0.02%-0.63%; 27 deaths; 14 studies). Among the 68 studies^{16,18,19,20,21,22,23,24,25,26,27,28,29,31,32,33,34,35,36,38,39,41,42,43,44,45,46,47,48,50,51,52,53,54,55,56,58,59,60,61,62,63,64,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,129} basing outcome assessment on medical records, the complication risk was 4.42% (95% CI, 3.49%-5.59%; 1005 complications) vs 8.91% (95% CI, 6.38%-12.31%; 3990 complication) among the 10 studies^{17,24,30,37,40,49,57,65,87,128} using administrative *ICD*-coded databases. Complication risks differed according to region but did not change over time ([Table 2](#)). In 17 studies^{16,18,19,23,31,33,40,43,46,50,54,56,58,59,70,81,129} wherein all patients were treated with

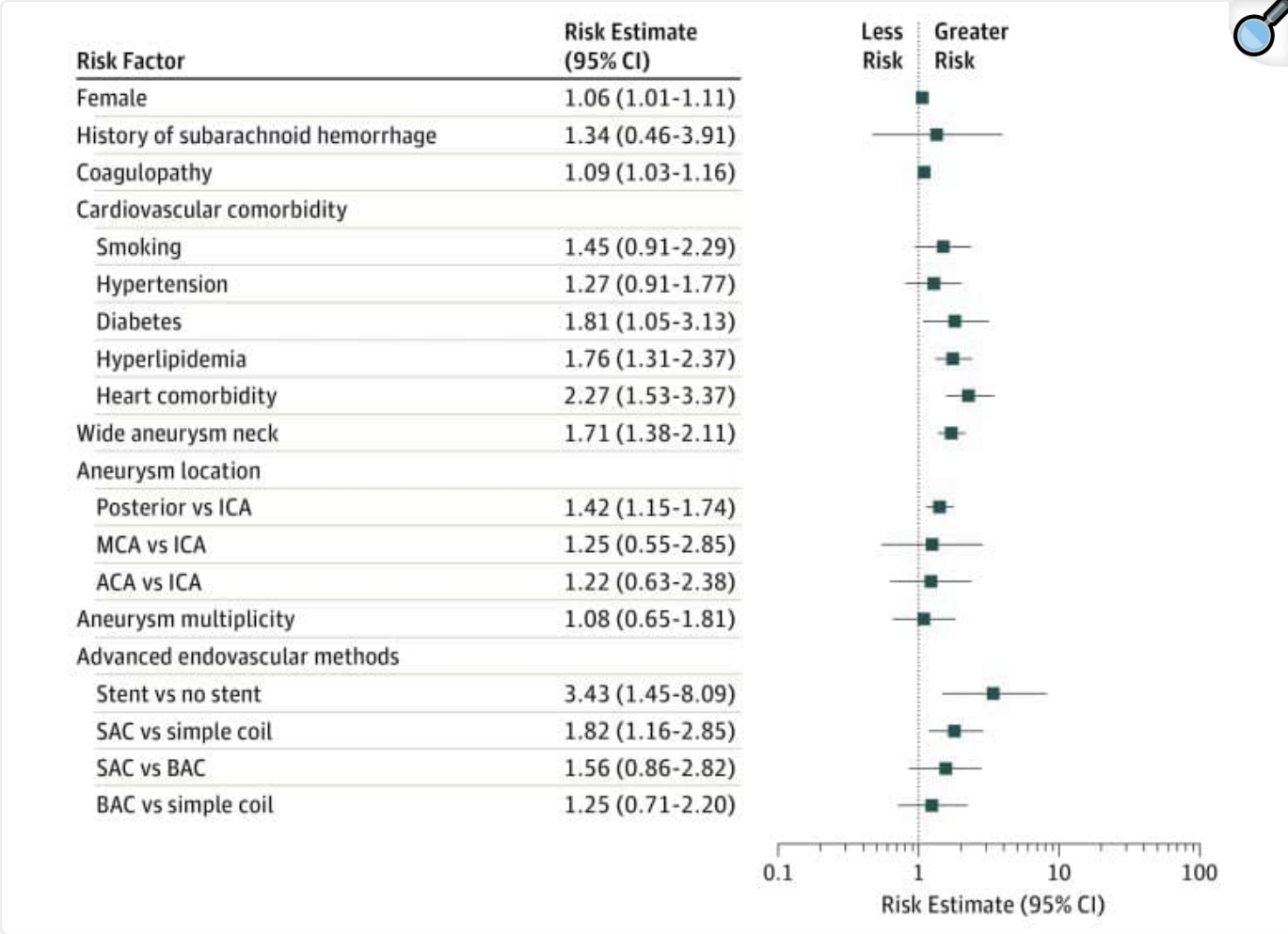
advanced endovascular methods, the pooled crude complication risk was 6.13% (95% CI, 4.29%-8.70%; 189 complications; [Table 2](#)) and the case-fatality rate was 0.43% (95% CI, 0.17%-1.10%; 14 deaths).

Meta-analyses of Risk Factor Data

Forty-

three [16,17,20,21,23,24,25,26,27,28,31,33,34,35,37,41,44,45,46,48,49,50,51,52,53,54,55,56,60,62,64,67,69,71,74,75,76,77,78,83,86,87,128](#) of 74 EVT studies (58%) reported on various risk factors for procedural clinical complications. An overview of all risk factors is given in eTable 5 in the [Supplement](#), and pooled risk factors are summarized in [Figure 1](#). Data on age, aneurysm size, and antiplatelet therapy could not be pooled (eFigures 2-4 in the [Supplement](#)). For 4809 female patients, the pooled OR for complications from 8 cohorts in 7 studies [20,21,26,28,45,46,62](#) was 1.06 (95% CI, 1.01-1.11; eFigure 5 in the [Supplement](#)). The existence of a coagulopathy and a history of SAH were not associated with an increased complication risk (eFigures 6 and 7 in the [Supplement](#)). The associations between cardiovascular risk factors and complications are summarized in eFigures 8-12 in the [Supplement](#). Patients with diabetes (4 cohorts from 3 studies [20,26,55](#); pooled OR, 1.81 [95% CI, 1.05-3.13]), hyperlipidemia (4 cohorts from 3 studies [20,26,49](#); pooled OR, 1.76 [95% CI, 1.31-2.37]), and cardiac comorbidity (3 cohorts from 2 studies [20,49](#); pooled OR, 2.27 [95% CI, 1.53-3.37]) were at increased risk of complications. A wide aneurysm neck (with a size of >4mm or a dome-to-neck ratio <1.5) was associated with an increased complication risk (5 cohorts from 4 studies [20,26,69,75](#); pooled OR, 1.71 [95% CI, 1.38-2.11]; eFigure 13 in the [Supplement](#)). Posterior circulation aneurysms were associated with an increased complication risk (6 studies [26,45,55,62,64,69](#); pooled OR, 1.42 [95% CI, 1.15-1.74]; eFigure 14 in the [Supplement](#)), but aneurysms localized at other locations were not. For aneurysm multiplicity, the pooled OR for complications was 1.08 ([95% CI, 0.65-1.81]; 3 cohorts from 2 studies [20,49](#); eFigure 15 in the [Supplement](#)). In total, 20 studies [16,20,21,23,26,33,34,44,45,46,48,50,52,53,55,62,75,78,87](#) reported on advanced endovascular methods (eFigure 16 in the [Supplement](#)). The use of stents was associated with an increased complication risk (compared with no stent use; 2 studies [26,62](#); pooled OR, 3.43 [95% CI, 1.45-8.09]), but no data on flow diverters could be meta-analyzed (eFigure 16 in the [Supplement](#)). Compared with standard coiling, the pooled OR for complications was 1.82 (5 studies [45,46,48,53,55](#); 95% CI, 1.16-2.85) in stent-assisted coiling and 1.25 (3 studies [45,55,75](#); 95% CI, 0.71-2.20) in balloon-assisted coiling.

Figure 1. Overview of the Association Between Various Patient, Aneurysm, and Treatment Factors and Risk of Procedural Clinical Complications From Endovascular Treatment.



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Abbreviations: ACA indicates anterior communicating artery; BAC, balloon-assisted coiling; ICA, internal carotid artery; MCA, middle cerebral artery; SAC, stent-assisted coiling.

Outcomes After NST

The pooled crude procedural risk from NST was 8.34% for any clinical complication (95% CI, 6.25%-11.10%; 54 studies^{[18,24,30,32,37,38,39,47,57,64,65,68,77,87,88,89,90,91,92,93,94,95,96,97,98,99,100,101,102,103,104,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119,120,121,122,123,124,125,126,127](#)}; 6501

complications; [Table 3](#)), 2.52% for ischemic complications (95% CI, 1.62%-3.91%; 33 studies [18,32,38,47,64,65,68,88,90,91,93,94,95,97,101,103,104,105,106,107,108,109,111,112,114,115,121,122,123,124,125,127](#); 509 complications), and 1.23% for hemorrhagic complications (95% CI, 0.71%-2.15%; 36 studies [18,32,38,47,64,65,68,88,90,92,93,94,99,102,103,104,105,106,107,108,109,110,112,113,114,115,119,120,121,122,123,125](#); 292 complications). The case-fatality rate was 0.10% (95% CI, 0.00%-0.20%; 49 studies [18,24,30,32,37,38,39,47,64,65,68,77,87,88,91,92,93,94,95,96,97,98,99,100,102,103,104,105,106,107,108,109,110,111,112,113,114,115,116,117,119,120,121,122,123,124,125,126,127](#); 156 deaths).

Table 3. Procedural Clinical Complication and Case-Fatality Rates From Neurosurgical Treatment for All Included Studies and by Subgroup.

Study Characteristic	Studies, No.	Events, No.	Patients, No.	Pooled Crude Risk, % (95% CI)	I^2 , %
All studies					
Procedural clinical complications	54	6501	34 614	8.34 (6.25-11.10)	99.0
Ischemic complications	33	509	12 691	2.52 (1.62-3.91)	94.3
Hemorrhagic complications	36	292	10 545	1.23 (0.71-2.15)	93.4
Case-fatality rate	49	156	24 901	0.10 (0.00-0.20)	91.4
High-quality studies					
Procedural clinical complications	11	303	4059	6.89 (3.80-12.16)	95.8
Case-fatality rate	11	5	4059	0.30 (0.00-0.94)	73.4
Procedural clinical complications					
Method of outcome assessment					
Medical records	43	761	10 265	6.43 (4.69-8.75)	94.2
ICD-coded administrative databases	11	5740	24 349	20.38 (14.69-27.56)	99.4
Standardized outcome scale ^a					
Yes	26	1906	12 590	8.27 (5.75-11.78)	97.4
No	28	4595	22 024	8.32 (5.29-12.86)	99.4
Region					
Europe	9	231	1314	14.77 (11.07-19.44)	94.2
North America	15	5851	25 247	18.41 (13.85-24.05)	99.1
Asia	29	407	7870	3.34 (2.88-6.50)	93.5

Study Characteristic	Studies, No.	Events, No.	Patients, No.	Pooled Crude Risk, % (95% CI)	I^2 , %
>1 Region	1	12	183	6.56 (2.97-10.14)	NA
Period					
2001-2007	18	3791	15 687	11.65 (7.62-17.41)	99.1
2008-2010	20	2091	14 382	8.16 (5.32-12.32)	98.6
2011-2014	15	602	4404	5.26 (2.57-10.44)	96.7
Period, case-fatality rate					
2001-2007	17	124	15 587	0.14 (0.03-0.65)	90.8
2008-2010	16	21	4769	0.04 (0.00-0.56)	88.3
2011-2014	15	11	4404	0.06 (0.00-0.80)	61.7

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Abbreviations: *ICD*, *International Classification of Diseases*; NA, not applicable.

^a Reporting neurological outcome using the modified Rankin Scale, the Glasgow Outcome Scale, or another standardized outcome scale.

Subgroup Analyses

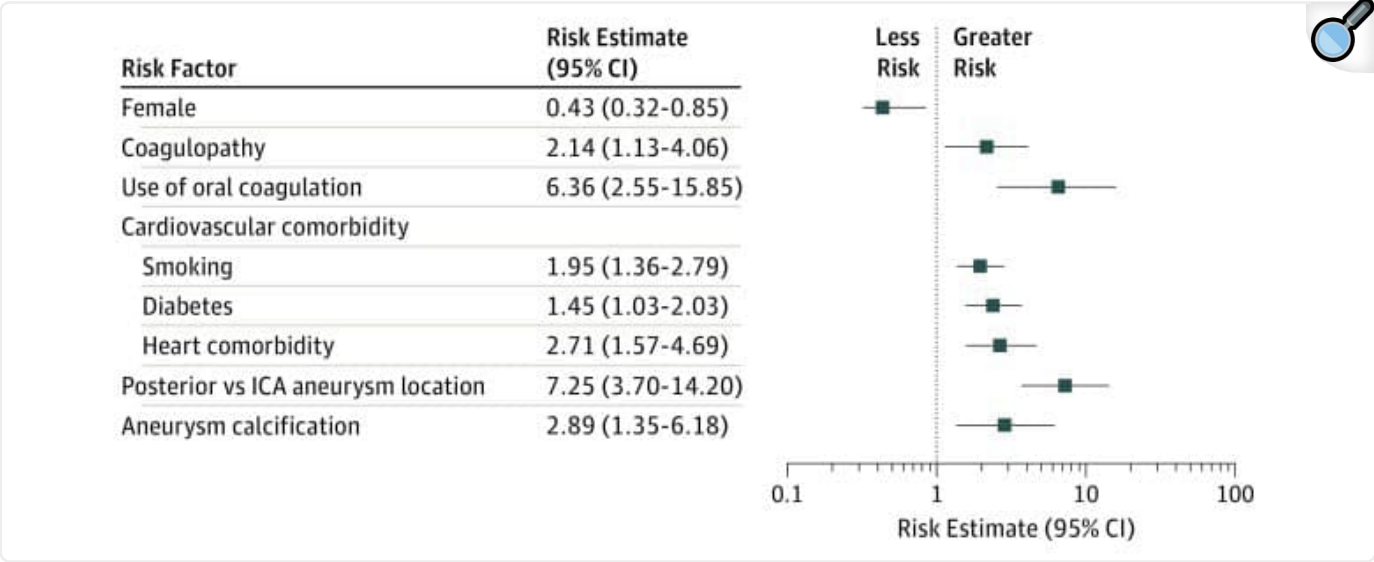
In 11 NST studies of high methodological quality, the complication risk was 6.89% (95% CI, 3.80%-12.16%; 303 complications; [Table 3](#)) and the case-fatality rate was 0.30% (95% CI, 0.00%-0.94%; 11 studies^{[64,77,95,102,110,112,113,118,119,120,127](#)}; 5 deaths). Among the 43 studies^{[18,32,38,39,47,64,68,77,88,91,92,93,94,95,97,98,99,100,102,103,104,105,106,107,108,109,110,111,112,113,114,116,117,118,119,120,121,122,123,124,125,126,127](#)} basing outcome assessment on medical records, the risk of complications was 6.43% (95% CI, 4.69%-8.75%; 761 complications) vs 20.38% (95% CI, 14.69%-27.56%; 5740 complications) among the 11 studies^{[30,37,57,65,87,89,90,96,101,115](#)} using *ICD*-coded databases. We found differences in complication risks according to region, with the highest complication risk in North America (15 studies^{[24,30,32,37,57,65,77,87,89,90,96,101,115,117,124](#)}; pooled crude risk 18.41% [95% CI, 13.85%-24.05%]; 5851 complications; including all 11 *ICD*-based studies^{[24,30,37,57,65,87,89,90,96,101,115](#)}). Complication risk decreased over time from 11.65% (95% CI, 7.62%-17.41%; 3791 complications; [Table 3](#)) in the period 2001 through 2007 (18

studies^{24,37,38,64,65,77,87,103,104,107,115,117,120,123,124,125,126,127}) to 5.26% (95% CI, 2.57%-10.44%; 602 complications) in the period 2011 through 2014 (15 studies^{18,30,32,39,88,93,97,100,105,110,111,112,113,114,118}). Case-fatality risks did not change over time ([Table 3](#)).

Meta-analyses of Risk Factor Data

Twenty-six^{24,37,65,77,87,89,91,95,96,99,101,102,104,106,108,110,112,113,115,116,117,118,119,120,125,126,127} of 54 NST studies (48%) reported on various risk factors for procedural clinical complications. A summary of all risk factors is given in eTable 5 in the [Supplement](#), and the pooled risk factors are summarized in [Figure 2](#). We did not pool data for aneurysm size and antiplatelet therapy (eFigures 3 and 4 in the [Supplement](#)). Ten studies reported on age: 7^{64,95,102,112,113,119,126} reported on age as a continuous variable (OR for complications per year increase, 1.02 [95% CI, 1.01-1.02]; eTable 5 in the [Supplement](#)) and 3^{91,99,120} reported different age categories (not pooled; eFigure 2 in the [Supplement](#)). For 3383 female patients, the pooled OR for complications from 10 studies^{91,95,99,102,110,112,113,117,119,120} was 0.43 (95% CI, 0.32-0.85; eFigure 5 in the [Supplement](#)). The risk of complications was increased in patients with a coagulopathy (2 studies^{101,115}; pooled OR, 2.14 [95% CI, 1.13-4.06]; eFigure 7 in the [Supplement](#)) and in those who used anticoagulation therapy (2 studies^{110,113}; pooled OR, 6.36 [95% CI, 2.55-15.85]; eFigure 3 in the [Supplement](#)). We found several cardiovascular risk factors to be associated with an increased complication risk (eFigures 8-12 in the [Supplement](#)): smoking (5 studies^{64,91,101,108,119}; pooled OR, 1.95 [95% CI, 1.36-2.79]), hypertension (5 studies^{64,91,101,108,119}; pooled OR, 1.45 [95% CI, 1.03-2.03]), diabetes (4 studies^{96,101,108,119}; pooled OR, 2.38 [95% CI, 1.54-3.67]), and congestive heart failure (2 studies^{115,119}; pooled OR, 2.71 [95% CI, 1.57-4.69]). Posterior circulation aneurysms were associated with an increased complication risk (pooled OR, 7.25 [95% CI, 3.70-14.20]; 3 studies^{64,104,112}; eFigure 14 in the [Supplement](#)). For aneurysm calcification, the pooled OR for complications was 2.89 (95% CI, 1.35-6.18; 2 studies^{77,126}; eFigure 17 in the [Supplement](#)).

Figure 2. Overview of the Association Between Various Patient, Aneurysm, and Treatment Factors and Risk of Procedural Clinical Complications From Neurosurgical Treatment.



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ICA indicates internal carotid artery.

Discussion

This systematic review and meta-analysis provides risk estimates of clinical complications and case-fatality rates for current preventive EVT and NST of saccular UIAs and identifies several patient-associated, aneurysm-associated, and treatment-associated risk factors for both treatments. We found substantial differences in complication risks according to region and method of outcome assessment. For NST, the complication risks decrease over time, but this is not true for EVT. The use of advanced endovascular methods is associated with an increased risk of clinical complications.

Comparison With Previous Studies

The pooled crude risks of clinical complications from EVT we found are in line with risks found in previous EVT reviews.^{9,130,135} In contrast, the NST complication risks we found were slightly higher than reported in previous reviews.^{10,130} A potential explanation for this discrepancy is that, in contrast with previous NST reviews, we did not restrict inclusion criteria

to studies reporting on permanent unfavorable outcomes but also included studies with nonpermanent complications and *ICD*-based administrative databases. In our subgroup analysis of studies basing outcome assessment on medical records, the NST complication risk was comparable with the risks found in previous NST reviews.^{10,130} The risks reported in *ICD*-coded databases were 2 to 3 times higher than in studies based on medical records. When interpreting these data, it should be kept in mind that all *ICD*-based studies were performed in North American hospitals, where correct listing of complications leads to higher reimbursement.¹³⁶ On the other hand, most of the studies basing their outcome on medical records were single-center or multicenter studies in which the surgeon or interventionist performed the retrospective analyses themselves, which may result in underestimating complication risks. One previous NST review also found that the complication risk was higher in a subgroup of North American studies.⁹ The much lower complication risk in studies originating from Asia has not been reported before. One explanation for this lower risk may be differences in how complications are defined and recorded. Another one is that a higher treatment volume per hospital or surgeon or interventionist leads to more experience with preventive aneurysm treatment, resulting in lower rates of complications.

Previous reviews on EVT and NST included studies published between 1990 and 2011 and reported that complication risks decreased for both EVT and NST over time. Although we did not find significant time trends for EVT and NST for clinical complications and case-fatality rates in the period between 2001 and 2014, we did find that the NST complication risk decreased more than 50% between the periods 2001 to 2007 and 2011 to 2014 and that case-fatality rates were in general lower in the overall period we studied (2001-2014) compared with earlier periods studied in previous reviews (a decrease from 1.5%-2.0% to <0.5%).^{9,10,130,131,137} Unfortunately, we were unable to perform time-trend analyses for standard coiling and advanced endovascular methods separately.

We found that stenting and stent-assisted coiling were associated with an increased complication risk, but balloon-assisted coiling was not. One previous EVT review⁹ found that the use of flow diverters was associated with a higher risk of an unfavorable outcome. More recently, several reviews of nonrandomized comparisons have been published on the outcome of various advanced endovascular methods, most of them confirming an increased complication risk for advanced endovascular methods compared with standard coiling.^{131,132,133,138,139,140} However, most studies included in these reviews^{131,132,133,138,139,140} included a mixture of saccular and nonsaccular aneurysms, making them unsuitable for our current review, which focused on saccular UIAs. For this reason, we also excluded 11 studies on flow diverters at the full-article screening stage (eTable 2 in the [Supplement](#)).

Other risk factors studied in previous EVT reviews^{9,130,135} are increasing age, sex, aneurysm diameter, aneurysm location, and aneurysm neck size. None of these factors were found to be

associated with an increased complication risk in these reviews, but risk factor data could only be studied from a small selection of articles.^{9,130,131,135} Separate reviews have been performed for subgroups of anterior and posterior circulation aneurysms.^{14,134} Previously identified risk factors for complications from NST are increasing age, aneurysm size, and posterior aneurysm location.^{10,132} None of the existing reviews included cardiovascular risk factors as potential determinants for procedural complications for EVT or NST.

Strengths

Our systematic review and meta-analysis has several strengths. To date, this work is the largest overview of UIA treatment outcomes, including data from more than 100 000 patients with treated saccular UIAs. This allowed us to study risk differences according to study design and region. Especially for EVT, we were now able to further explore the association between the use of various advanced endovascular methods and risk of complications. A second strength is that we also studied risk factor data in detail. This enabled us to add several new aneurysm- and treatment-related risk factors to the already known risk factors from the literature. In addition, this is the first joint endovascular and neurosurgical meta-analysis to give a complete overview of the impact of various cardiovascular risk factors on both treatments.

Limitations

Some limitations need to be addressed. First, the complication risks for EVT and NST should not be compared because of the nonrandomized nature of the included studies, which makes them prone to various sources of bias, such as selection bias. So far, only 1 randomized clinical trial¹⁵ has been published on EVT vs NST in patients with saccular UIAs, which assessed permanent morbidity at 1 year as a secondary outcome. Second, only approximately 20% of the included studies were of high methodological quality. Third, outcome definitions were very heterogeneous across studies, and we were not able to disentangle transient and permanent clinical complications. This heterogeneity underscores the need of cautious interpretation of our meta-analyses of all clinical complications combined. Fourth, a part of the included studies reported very limited data on patient and aneurysm characteristics. As a result, we were limited in our analyses. Fifth, the finding that the complication risk is higher in subgroups of patients treated by NST or advanced endovascular methods may be a reflection of the complex nature of aneurysms treated by such treatment modalities. Finally, in our meta-analysis of risk factors for procedural clinical complications, we relied on the definitions and categorizations of risk factors and reference groups given in the original articles. For several risk factors, such as age and advanced endovascular methods, this meant that studies were noncomparable and could not be pooled or only a subset could be pooled.

Conclusions

This review provides precise estimates of procedural clinical complications and case-fatality rates from preventive EVT or NST of UIAs. The complication risk varies according to several patient-associated, aneurysm-associated, and treatment-associated risk factors. Most published observational data on preventive UIA treatment remain of poor methodological quality, with sensitivity particularly dependent on detailed and standardized recording of procedural clinical complications, method of outcome assessment and region. For clinical practice, the data from this study can be used to estimate the procedural complication risk from preventive UIA treatment according to patient-associated, aneurysm-associated, and treatment-associated characteristics, which need to be balanced against the risk of rupture when preventive aneurysm treatment is considered.

Through future research, our work can be further extended by meta-analysis of individual patient data from studies of high methodological quality. We underscore the need for detailed and standardized recording of clinical complications and treatment risk factors in a prospective setting to allow for multivariable analyses assessing the independent contribution of the different risk factors. With such data available, scores can be developed to prognosticate individualized procedural complication risks according to each person's risk factor profile.

Supplement.

eTable 1. Detailed search query

eTable 2. Excluded articles at full-text assessment

eTable 3. Baseline characteristics for included studies on endovascular treatment (EVT)

eTable 4. Baseline characteristics for included studies on neurosurgical treatment (NST)

eTable 5. Included studies reporting risk factor data

eFigure 1. Flowchart

eFigure 2. Forest plot of the association between age and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 3. Forest plot of the association between use of antiplatelet therapy (APT) and/or anticoagulation therapy (ACT) and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 4. Forest plot of the associations between aneurysm size and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 5. Forest plot of the association between female sex and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 6. Forest plot of the associations between history of SAH and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 7. Forest plot of the associations between coagulopathy and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 8. Forest plot of the associations between smoking and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 9. Summary forest plot of the associations between hypertension and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 10. Forest plot of the associations between diabetes and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 11. Forest plot of the associations between hyperlipidemia and risk of procedural treatment complications following EVT

eFigure 12. Forest plot of the associations between heart comorbidity and risk of procedural clinical complications following EVT (A) and NST(B)

eFigure 13. Forest plot of the association between aneurysm neck size and risk of procedural clinical complications following EVT

eFigure 14. Forest plot of the associations between aneurysm location and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 15. Forest plot of the association between aneurysm multiplicity and risk of procedural clinical complications following EVT

eFigure 16. Forest plot of the association between use of various advanced endovascular methods and risk of procedural clinical complications

eFigure 17. Forest plot of the association between aneurysm calcification and risk of procedural clinical complications following NST

eReferences. Excluded articles

[Click here for additional data file.](#) (490.9KB, pdf)

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

Supplement.

eTable 1. Detailed search query

eTable 2. Excluded articles at full-text assessment

eTable 3. Baseline characteristics for included studies on endovascular treatment (EVT)

eTable 4. Baseline characteristics for included studies on neurosurgical treatment (NST)

eTable 5. Included studies reporting risk factor data

eFigure 1. Flowchart

eFigure 2. Forest plot of the association between age and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 3. Forest plot of the association between use of antiplatelet therapy (APT) and/or anticoagulation therapy (ACT) and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 4. Forest plot of the associations between aneurysm size and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 5. Forest plot of the association between female sex and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 6. Forest plot of the associations between history of SAH and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 7. Forest plot of the associations between coagulopathy and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 8. Forest plot of the associations between smoking and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 9. Summary forest plot of the associations between hypertension and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 10. Forest plot of the associations between diabetes and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 11. Forest plot of the associations between hyperlipidemia and risk of procedural treatment complications following EVT

eFigure 12. Forest plot of the associations between heart comorbidity and risk of procedural clinical complications following EVT (A) and NST(B)

eFigure 13. Forest plot of the association between aneurysm neck size and risk of procedural clinical complications following EVT

eFigure 14. Forest plot of the associations between aneurysm location and risk of procedural clinical complications following EVT (A) and NST (B)

eFigure 15. Forest plot of the association between aneurysm multiplicity and risk of procedural clinical complications following EVT

eFigure 16. Forest plot of the association between use of various advanced endovascular methods and risk of procedural clinical complications

eFigure 17. Forest plot of the association between aneurysm calcification and risk of procedural clinical complications following NST

eReferences. Excluded articles

[Click here for additional data file.](#) (490.9KB, pdf)